PH-101: Mathematical Physics - I (Theory)

Lecture Note 1 ver: 1.1.0

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1. Introduction to Ordinary Differential Equations

We now introduce the fundamental concept that we are going to study in our class, an Ordinary Differential Equation (ODE). INFORMALLY speaking an ODE is an equation describing the dynamics of any object. Let us elaborate (informally) so that we have a better understanding of the concept. Consider a physical quantity x which changes w.r.t time t and we want to study this change. We would first need to find out what quantities affect this change. For our example let us take two other physical quantities y and z along with x itself that affect this change. We know from basic calculus that we can represent the instantaneous rate of change of x with respect to (from now on we will write w.r.t) time t by the derivative of x w.r.t t, i.e. $\frac{\mathrm{d}}{\mathrm{d}t}x$. The quantity x along with its derivative w.r.t t represents the dynamics of x and if the relation between this derivative and the quantities x, y, and z can be represented by an equation that equation will be an ODE. The equation can be of any form, it just needs to include the derivative and the other quantities(as per requirement), for example

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = af(y) + bg(x, z) \tag{1}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = af(y) + bg(x, y) + cz \tag{2}$$

$$\left(\frac{\mathrm{d}}{\mathrm{d}t}x(t)\right)^2 = af(y) + bg(x) + cw(z) \tag{3}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f(x, y, z) \tag{4}$$

All the equations listed above are ODEs. Now the equations can also involve derivatives of other physical quantities and even higher order derivatives of these quantities also, they would still be considered ODEs, for example

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = af\left(y, \frac{\mathrm{d}}{\mathrm{d}t}y\right) + bg(z, x) + cw\left(\frac{\mathrm{d}}{\mathrm{d}t}z\right) \tag{5}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f\left(x, y, z, \frac{\mathrm{d}}{\mathrm{d}t}z\right) \tag{6}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f\left(x, y, z, \frac{\mathrm{d}}{\mathrm{d}t}y, \frac{\mathrm{d}}{\mathrm{d}t}z\right) \tag{7}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}x(t) + \left(\frac{\mathrm{d}^3}{\mathrm{d}t^3}x(t)\right)^4 + a\frac{\mathrm{d}}{\mathrm{d}t}x(t) = f\left(x, y, z, \frac{\mathrm{d}}{\mathrm{d}t}z, \frac{\mathrm{d}^2}{\mathrm{d}t^2}y\right) \tag{8}$$

Let us now formally define Ordinary Differential Equations (ODEs).

Definition (Ordinary Differential Equation): Let us assume $I \underset{\text{open}}{\subseteq} \mathbb{R}$ is an open interval in $\mathbb{R}$, $t \in I$ is an independent scalar variable with $x : I \to \mathbb{R}^n$ defined as an m-times continuously differentiable function on I and $D \underset{\text{open}}{\subseteq} \mathbb{R}^{1+n(m+1)}$ is an open subset of $\mathbb{R}^{1+n(m+1)}$ with $F : D \to \mathbb{R}^n$ defined as a continuous function on D. An Ordinary Differential Equation (ODE) is a relation of the form

$$F\big(t, x(t), \dot{x}(t), ..., x^{(m)}(t)\big) = 0 \tag{9}$$

We consider $y : J \to \mathbb{R}^n$ an m-times continuously differentiable function defined on the open interval $J \underset{\text{open}}{\subseteq} I$ to be a *classical solution* of Equation 9 if $\big(t, y(t), \dot{y}(t), ..., y^{(m)}(t)\big) \in D$ and Equation 9 holds for all $t \in J$.

We now have a formal definition of an ODE and it's solution. Now, let us study the various types of ODEs.

We begin by introducing some tools for labeling ODEs like the *order of an ODE* and *the degree of an ODE*. The order of the highest-order derivative of the dependent variable w.r.t the independent variable involved in an ODE is defined as the *order of the ODE*. The degree to which that highest-order derivative term is raised to in the ODE is defined as the *degree of the ODE*.

In the definition of an ODE we mention a function F and define an ODE based on a relation involving this function F. We label an ODE as an *implicit ordinary differential equation* if the relation describing the ODE is of the form

$$F\big(t, x(t), \dot{x}(t), ..., x^{(m)}(t)\big) = 0 \tag{10}$$

where, t and x are as defined in the definition.

We label the ODE as an *explicit ordinary differential equation* if the relation describing the ODE can be written in the form

$$\frac{\mathrm{d}^m}{\mathrm{d}t^m} x(t) = \widetilde{F}\big(t, x(t), \dot{x}(t), ..., x^{(m-1)}(t)\big) \tag{11}$$

where, t and x are as defined earlier and $\widetilde{F} : \widetilde{D} \to \mathbb{R}^n$ and $\widetilde{D} \underset{\text{open}}{\subseteq} \mathbb{R}^{1+nm}$.

Now the definition for an ODE we provided earlier can also be simplified and in many textbooks this simplified version is provided. In the simplified version, the derivatives of the variable x w.r.t t are included as components of x and therefore the higher order derivatives are converted to a first order derivative of the new variable x. We will state that version too for completeness. We present such a version of the definition of an ODE from a popular book for Ordinary Differential Equations by Jack K. Hale [1].

Definition (Ordinary Differential Equation): Let us assume $I \underset{\text{open}}{\subseteq} \mathbb{R}$ is an open interval in $\mathbb{R}$, $t \in I$ is an independent scalar variable with $x : I \to \mathbb{R}^n$ defined as a continuously differentiable function on I and $D \underset{\text{open}}{\subseteq} \mathbb{R}^{n+1}$ is an open subset of $\mathbb{R}^{n+1}$ with $f : D \to \mathbb{R}^n$ defined as a continuous function on D. An Ordinary Differential Equation (ODE) is a relation of the form

$$\frac{\mathrm{d}}{\mathrm{d}t} x(t) = f(t, x(t)) \tag{12}$$

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We consider $y : J \to \mathbb{R}^n$ a continuously differentiable function defined on the open interval $J \underset{\text{open}}{\subseteq} I$ to be a *classical solution* of Equation 12 if $(t, y(t)) \in D$ and Equation 12 holds for all $t \in J$.

We now define a *Linear Ordinary Differential Equation* as an ordinary differential equation where the differential operators involved are linear in the dependent variables and their derivatives. In case of Equation 9 and Equation 12 we need F and f to be a linear combination of x and its derivatives to be considered a linear ODE. Now, there are two types of linear ODEs, the first called *homogeneous* where the linear combinarion of the dependent variable and its derivatives, that forms the linear ODE, is equal to zero. It is *non-homogeneous* if it is not equal to zero.

Here are some examples of homogeneous linear ODEs

$$a\left(\frac{\mathrm{d}}{\mathrm{d}x}y(x)\right) + g(x)\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}y(x)\right) + k(x)\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + p(x)y(x) = 0 \tag{13}$$

$$a\left(\frac{\mathrm{d}}{\mathrm{d}x}y(x)\right) + b\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}y(x)\right) + b\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + dy(x) = 0 \tag{14}$$

$$g(x)\left(\frac{\mathrm{d}^4}{\mathrm{d}x^4}y(x)\right) + k(x)\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + ay(x) = 0 \tag{15}$$

We here also list some non-homogeneous linear ODEs

$$a\left(\frac{\mathrm{d}}{\mathrm{d}x}y(x)\right) + g(x)\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}y(x)\right) + k(x)\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + p(x)y(x) = z(x) \tag{16}$$

$$a\left(\frac{\mathrm{d}}{\mathrm{d}x}y(x)\right) + b\left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}y(x)\right) + b\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + dy(x) = p \tag{17}$$

$$g(x)\left(\frac{\mathrm{d}^4}{\mathrm{d}x^4}y(x)\right) + k(x)\left(\frac{\mathrm{d}^3}{\mathrm{d}x^3}y(x)\right) + ay(x) = l(x) \tag{18}$$

Bibliography

[1] J. K. Hale, *Ordinary differential equations*. Courier Corporation, 2009.